Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

Show Ip route ospf. 2

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129 2

Does R2 show up as an OSPF neighbor on R1? Yes 2

Does R2 show up as an OSPF neighbor on R3? No 2

What does this information tell you?

R2's S0/0/1 interface is still set to passive, so it does not send OSPF hello packets to R3 and cannot form a neighbor adjacency with R3. However, R2's S0/0/0 is not passive, so R1 can still see R2 as a neighbor. R3 has to reach the 192.168.2.0/24 network indirectly through R1, which is why the cost is higher (64 + 64 + 1 = 129). 2

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1
R2(config-router)# no passive-interface s0/0/1 2

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

The accumulated cost is 65. It is calculated by adding the cost of R3's S0/0/1 interface (64) and R2's G0/0 interface (1), so 64 + 1 = 65. This is now the preferred route because it has a lower cost than going through R1. 2

Is R2 listed as an OSPF neighbor to R3? Yes 2

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

The default reference bandwidth is 100 Mbps, which means any link that is 100 Mbps or faster such as Fast Ethernet, Gigabit Ethernet and 10G Ethernet, these all get the same OSPF cost of 1. This makes it impossible for OSPF to distinguish between these different link speeds. Changing the reference bandwidth to a higher value allows OSPF to assign different costs to different speed links, so routing decisions are more accurate.

2

Step 2: Change the bandwidth for an interface.

g. Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The cost to 192.168.3.0/24 is 782, calculated as R1 S0/0/1 cost (781) + R3 G0/0 cost (1) = 782.
The cost to 192.168.23.0/30 is 845, calculated as R1 S0/0/1 cost (781) + R3 S0/0/1 cost (64) = 845.

2

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

The new accumulated cost is 1562. After setting bandwidth 128 on all serial interfaces, each serial link now has a cost of 781. The route to the 192.168.23.0/24 network travels over two serial links, so the cost is 781 + 781 = 1562.

2

Step 3: Change the route cost.

c. Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF always chooses the route with the lowest total cost. The direct path from R1 to R3 via S0/0/1 now has a manually set cost of 1565, so the total cost to reach 192.168.3.0/24 directly would be 1565 + 1 = 1566. The path through R2 costs R1 S0/0/0 (781) + R2 S0/0/1 (781) + R3 G0/0 (1) = 1563, which is lower than 1566. So OSPF prefers the path through R2.

4

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   The router ID uniquely identifies each router in the OSPF domain and is used in the DR/BDR election process. If the router ID is based on a physical interface address and that interface goes down, the router ID could change, which causes instability in the OSPF network. By manually setting the router ID using the router-id command or a loopback interface, we can ensure the router ID remains stable and predictable.

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2. Why is the DR/BDR election process not a concern in this lab?

   The DR/BDR election only takes place on multi-access networks such as Ethernet. In this lab, the connections between routers are all point-to-point serial links, so no DR or BDR election is needed on those interfaces. The GigabitEthernet LAN interfaces each only have one router connected, so there is no other OSPF router to elect a DR/BDR with.

   4

3. Why would you want to set an OSPF interface to passive?

   Setting a LAN interface to passive prevents the router from sending unnecessary OSPF hello packets out to end devices like PCs, which do not need to receive routing protocol traffic. This saves bandwidth on the LAN segment and also improves security by not exposing OSPF routing information to devices that do not need it.

   4

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   The use of OSPF areas makes the protocol hierarchical. By dividing a large network into multiple areas, OSPF can reduce the size of the routing table and the amount of link-state advertisement traffic, which allows it to scale to very large networks.

   4

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   network 10.0.0.0 0.255.255.255 area 0.

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6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   The EIGRP route will be used. When a router receives routes to the same network from different routing protocols, it uses the administrative distance (AD) to decide which route to prefer. EIGRP has an AD of 90, OSPF has an AD of 110, and RIP has an AD of 120. Since EIGRP has the lowest AD, its route is considered the most trustworthy and will be installed in the routing table.

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